CLASS 9 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

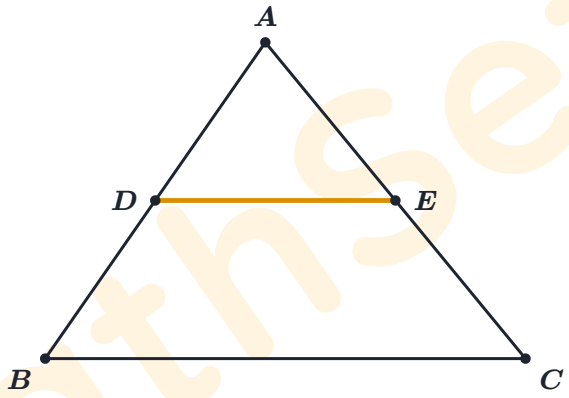
Chapter 12: Mid-point and Its Converse

Topic-wise short notes with solved examples

The mid-point theorem, its converse and the equal intercept theorem, with their use in triangles, trapeziums and other quadrilaterals.

1 Mid-point theorem

- **Theorem**: the line segment joining the **mid-points of two sides** of a triangle is **parallel to the third side** and **equal to half of it**.
- If D and E are the mid-points of AB and AC, then $DE \parallel BC$ and $DE = \frac{1}{2}BC$.



D and E are mid-points: DE is parallel to BC and DE = ½ BC

- **Proof idea**: produce DE to F so that $EF = DE$ and join CF. $\triangle ADE \cong \triangle CFE$ (S.A.S.), so $CF = AD = DB$ and $CF \parallel AB$. Then DBCF is a parallelogram, so $DF \parallel BC$ and $DF = BC$, giving $DE = \frac{1}{2}BC$.

Example 1.1 In triangle ABC, D and E are the mid-points of AB and AC. If $BC = 14$ cm and $\angle B = 65°$, find DE and $\angle ADE$.

SOLUTION

$DE = \frac{1}{2} \times 14$. Since $DE \parallel BC$, $\angle ADE = \angle B$ (corresponding angles).

$DE = 7$ cm, $\angle ADE = 65°$

2 Converse of the mid-point theorem

- **Theorem**: the line drawn through the **mid-point of one side** of a triangle, **parallel to another side**, **bisects the third side**.
- If D is the mid-point of AB and $DE \parallel BC$ meets AC at E, then E is the mid-point of AC, and $DE = \frac{1}{2}BC$.

Example 2.1 In triangle ABC, D is the mid-point of AB and DE is parallel to BC, meeting AC at E. If $AC = 9$ cm and $BC = 10$ cm, find AE and DE.

SOLUTION

E is the mid-point of AC.

$AE = 4.5$ cm, $DE = 5$ cm

3 The triangle of mid-points

- Let D, E, F be the mid-points of BC, CA, AB of triangle ABC.
- $EF \parallel BC, FD \parallel CA, DE \parallel AB$, and each is half the side it is parallel to.
- The four triangles AFE, FBD, EDC and DEF are **congruent**. Each has $\frac{1}{4}$ of the area of triangle ABC.
- Perimeter of triangle DEF $= \frac{1}{2} \times$ perimeter of triangle ABC.
- BDEF, DCEF and AFDE are parallelograms.

Example 3.1 The sides of a triangle are 8 cm, 10 cm and 12 cm. Find the perimeter of the triangle formed by joining the mid-points of its sides.

SOLUTION

The sides are 4 cm, 5 cm and 6 cm.

15 cm

4 Quadrilaterals

- **The figure formed by joining the mid-points of the sides of any quadrilateral, in order, is a parallelogram.**
- Reason: if P, Q, R, S are the mid-points of AB, BC, CD, DA, then PQ and SR are both parallel to AC and equal to $\frac{1}{2}AC$.

Original quadrilateral	Mid-point figure
Any quadrilateral	Parallelogram
Rectangle	Rhombus
Rhombus	Rectangle
Square	Square
Diagonals equal	Rhombus
Diagonals perpendicular	Rectangle

Example 4.1 The diagonals of a quadrilateral are 10 cm and 16 cm. Find the perimeter of the figure formed by joining the mid-points of its sides.

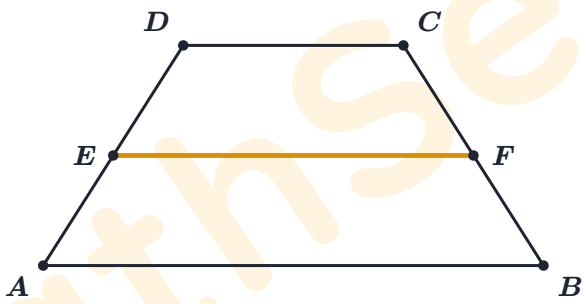
SOLUTION

Two sides are $\frac{1}{2} \times 10 = 5$ cm and two are $\frac{1}{2} \times 16 = 8$ cm.

26 cm

5 Trapezium

- In trapezium ABCD with $AB \parallel DC$, let E and F be the mid-points of the non-parallel sides AD and BC.
- $EF \parallel AB \parallel DC$ and $EF = \frac{1}{2}(AB + DC)$.
- The segment joining the mid-points of the **diagonals** is parallel to the parallel sides and equals $\frac{1}{2}(AB - DC)$, where $AB > DC$.



EF joins the mid-points of AD and BC: EF = ½ (AB + DC)

Example 5.1 In trapezium ABCD, $AB \parallel DC$, $AB = 18$ cm and $DC = 10$ cm. E and F are the mid-points of AD and BC. Find EF.

SOLUTION

$\frac{1}{2}(18 + 10)$

14 cm

Example 5.2 In the same trapezium, find the distance between the mid-points of the diagonals.

SOLUTION

$\frac{1}{2}(18 - 10)$

4 cm

6 Equal intercept theorem

- **Theorem**: if a transversal makes **equal intercepts** on three or more parallel lines, then **any other transversal** cutting them also makes equal intercepts.

- If $l \parallel m \parallel n$ and one transversal has $AB = BC$, another transversal has $PQ = QR$.
- This theorem is used to divide a line segment into any number of equal parts.

Example 6.1 Three parallel lines cut a transversal at A, B, C with $AB = BC = 3$ cm. They cut another transversal at P, Q, R with $PQ = 4$ cm. Find PR.

SOLUTION

$QR = PQ = 4$ cm

$PR = 8$ **cm**

Example 6.2 ABCD is a parallelogram. E is the mid-point of AB and F is the mid-point of CD. Prove that AF and CE trisect the diagonal BD.

SOLUTION

Let AF and CE meet BD at P and Q. AECF is a parallelogram ($AE = CF$ and $AE \parallel CF$), so $AF \parallel EC$.

In $\triangle ABP$, E is the mid-point of AB and $EQ \parallel AP$, so Q is the mid-point of BP: $BQ = QP$. Similarly in $\triangle DQC$, $DP = PQ$.

$DP = PQ = QB$

7 Common mistakes

- Using the mid-point theorem when only one of the points is a mid-point. Then the converse is needed, together with a parallel line.
- Writing $DE = BC$ instead of $DE = \frac{1}{2}BC$.
- In a trapezium, taking the mid-points of the parallel sides instead of the non-parallel sides.
- Forgetting to state both parts of the theorem (parallel **and** half).
- Assuming the mid-point figure of a quadrilateral is a rectangle or rhombus without checking the diagonals.

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